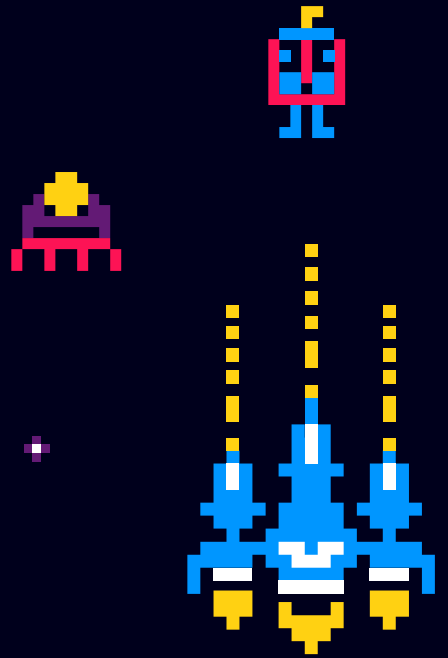
ElasticFR #101

FLEET INVADERS



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EXTRACT TRANSFORM SEARCH

About us

PIETRO MELE

Software Engineer

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Search consultant

Our company

Experts in Search technologies

Integrators of Elasticsearch, OpenSearch et Solr

Consulting and formations

Editor of a2 : Solution Search for E-Commerce

Editor of all.site - Collaborative Search Engine

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Telemetry collection

The goal is to collect everything:

- Logs
- Metrics
- Uptime

The fragmented arsenal:

- Filebeat
- Metricbeat
- Heartbeat

L'Arsenal Fragmenté :

The problems

Operational overload

- Multiple beat for machine, one for each purpose
- Configuration by YAML: multiple files for each service.

When we have to configure this on multiple machines ....

GAME
OVER

GAME ON!

Introducing Fleet, agents control center

Fleet

- Roll-in, roll-out and upgrade of agents centralized and make easy
- Come with an UI in Kibana
- Need to configure at least one fleet server to use Fleet on premise

Agents

- Allows to collects
  - logs
  - metrics
  - uptime

- Not only monitoring, but also security
- One agent per host can monitor multiple services

Integrations

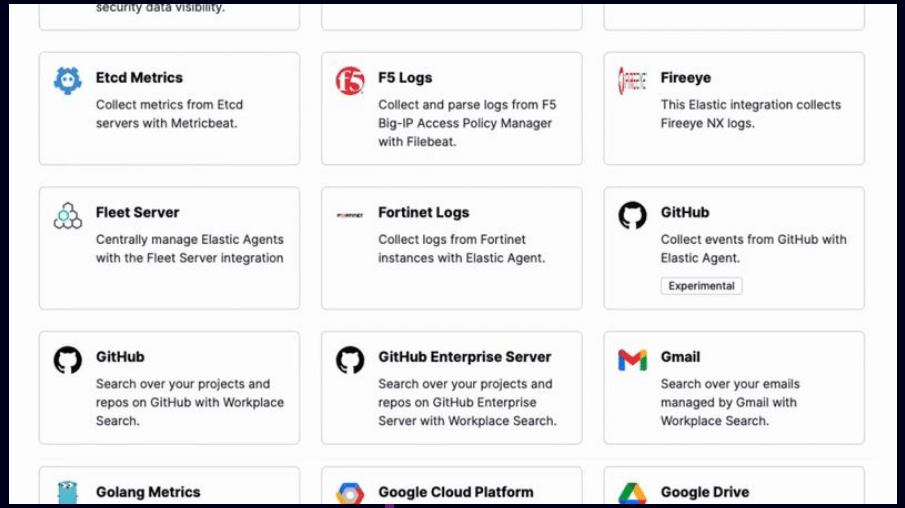
- The agent can be furnished with multiple integrations.
- There are more than 400 integrations
- Compliant with the ECS
- Integrations comes with useful assets:
  - pipelines
  - dashboards
  - templates
  - .....

Integrations

SEARCH

SECURITY

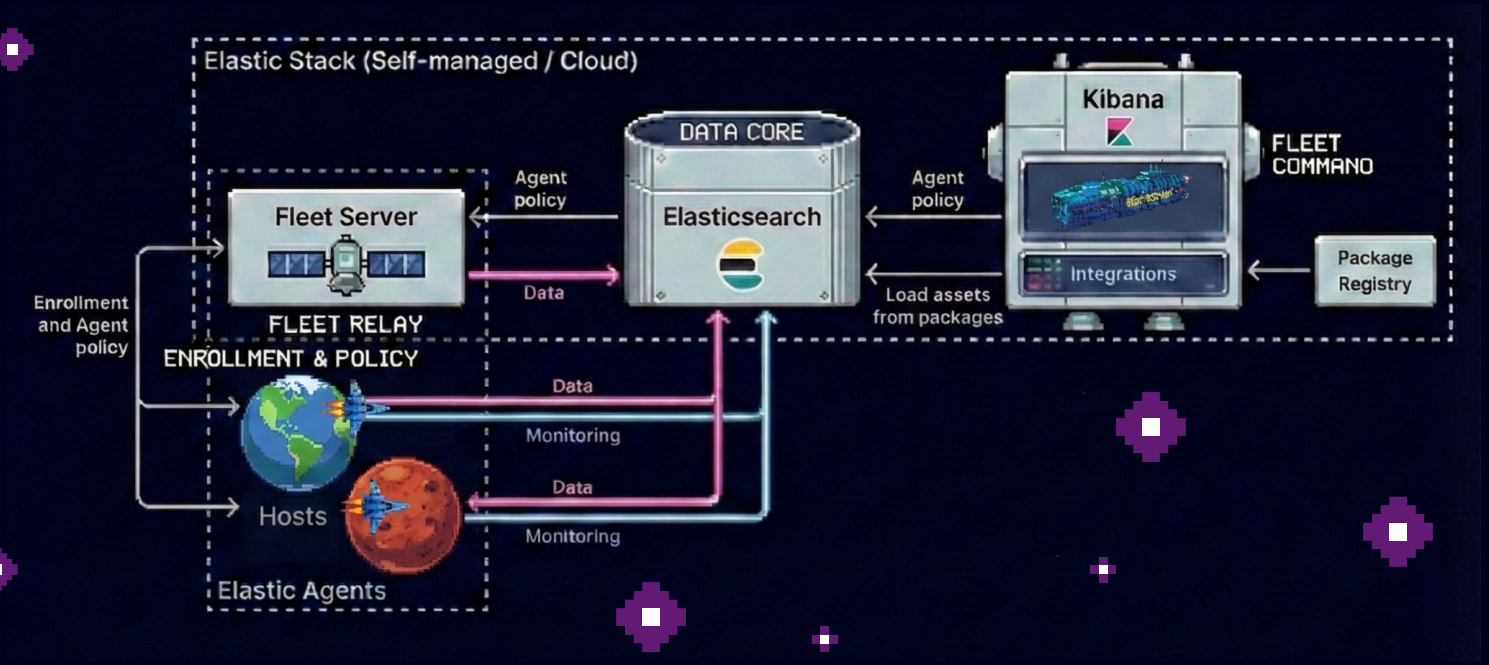
OBSERVABILITY



Policies

- Specify which integrations to run
- Handle the settings for each integration
- One policy can be applied to multiple agents
- Two or more policies can't be applied to the same agent

Architecture Overview



Enrollment process

01 Create a policy

02 Generate an enrollment token

03 Download and unpack the agent

04 Enroll the agent

05 Wait for data

DEMO

Hopefully without any demo effect

Air-gapped environment

- Fleet need access to the Elastic Package Registry and the Elastic Artifact Registry
- xpack.fleet.isAirGapped: true
- Can use a proxy access
- Can host the registries
- EPR is available as a Docker image
- EAR is an HTTP file server with binaries

Stand alone agent

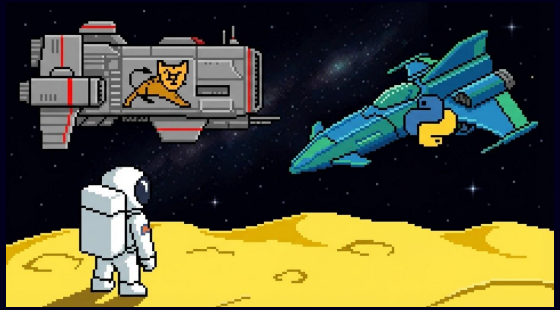
- An agent can run without a Fleet server
- All the settings in the elastic-agent.yml file
- The policy creates the content of the file
- Upgrade is manual

Installation : archive or not archive ?

- Install via package or archive
- Package way is easier
- Archive way is more flexible (auto update)

What kind of integrations

- Many OOB integrations
- Customized pipelines and index templates
- Dashboards
- Security and observability features



One limit - for now

- One agent can only be attached to a single policy
- It's not possible (yet ?) to have a "core" policy (system) and then specific application policies

Custom integrations

- Generic integrations (Custom logs)
- AI assistant is here to help create a custom integration from a sample of the data

Agent migration

- When creating a new cluster, an agent can be enrolled into it with some simple operations

Takeaways

Telemetry collection is evolving

Fleet brought this to a new level:

- It consolidate multiple collectors into one Elastic Agent.
- It centralize the control to a single interface in Kibana.
- It eases scaling

Works on cloud, on premise and in air-gapped environments

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MERCI !

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